

Smart Chunking & Retrieval System

Terminal Usage Guide

This guide explains how to use the CLI tools for the Smart Chunking & Retrieval System from your terminal. The system consists of a main execution script (`main.py`) and a test suite (`test_chunking.py`).

Prerequisites: Ensure you have Python 3 installed. If you intend to process PDF files (like the provided `paper.pdf`), you must install the required dependency:

```
pip install pdfplumber
```

1. Using the Main CLI (`main.py`)

The `main.py` script is the primary entry point to analyse documents and retrieve relevant chunks based on a query.

Running the Built-in Demo

To quickly test the system without providing any files, run the built-in demo. This uses a hardcoded text about Machine Learning and a predefined query.

```
python main.py --demo
```

Basic Document Retrieval

To query a specific document, use the `--doc` and `--query` arguments. The system supports `.txt` , `.json` , and `.pdf` files.

```
python main.py --doc report.txt --query "What is unsupervised learning?"
```

```
python main.py --doc data.json --query "Explain the Transformer architecture"
```

Advanced Configuration

You can fine-tune the chunking parameters to optimise retrieval for specific types of documents:

- `--top-k`: Number of top chunks to retrieve (default: 5).
- `--semantic-threshold`: Cosine similarity drop threshold for semantic chunking (default: 0.15).
- `--overlap-window`: Sentence window size for overlap chunking (default: 6).
- `--overlap-size`: Number of overlapping sentences between windows (default: 2).

```
python main.py --doc paper.pdf --query "Adversarial attacks" --top-k 3 --overlap-window 8 --semantic-threshold 0.2
```

JSON Output

If you need to pipe the output to another program or save the raw data, use the `--json-output` flag to dump the full result dictionary to stdout.

```
python main.py --doc report.txt --query "Healthcare applications" --json-output
```

2. Running the Test Suite (`test_chunking.py`)

The system includes a comprehensive test suite to verify the core algorithms (tokenisation, TF-IDF, chunking strategies, and scoring).

To run the tests, simply execute the test script from the terminal:

```
python test_chunking.py
```

This will output a summary of passed and failed tests directly to your console, ensuring all core functionalities are working as expected.